

Stability Theorem for Causal SISO LTI Systems

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Definition of Bounded Input/Output

A causal SISO LTI system's input $u(t)$ is **bounded** if

$$\|u\|_{\infty} := \sup_{t \in [0, \infty)} |u(t)| < \infty$$

Similarly, its output $y(t)$ is **bounded** if

$$\|y\|_{\infty} := \sup_{t \in [0, \infty)} |y(t)| < \infty$$

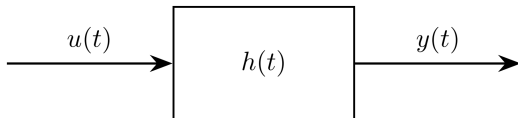
Examples

$u(t) = t$ is not bounded because $\|u\|_{\infty} = \sup_{t \in [0, \infty)} |t| = \infty$

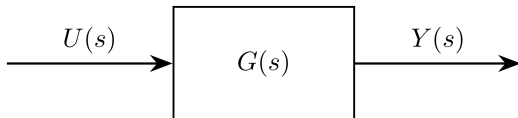
$u(t) = \sin t$ is bounded because $\|u\|_{\infty} = \sup_{t \in [0, \infty)} |\sin t| = 1 < \infty$

Definition of Stability

Consider a causal SISO LTI system shown in t domain:



or equivalently (recall $G(s) := \mathcal{L}[h(t)]$) in s domain:



We say this system is bounded input bounded output (BIBO) stable, or internally stable, or stable, if

bounded input implies bounded output, i.e., $\|u\|_{\infty} < \infty \Rightarrow \|y\|_{\infty} < \infty$

Stability Theorem: Statement

For a causal SISO LTI system with impulse response function $h(t)$ and rational transfer function $G(s)$, the following statements are equivalent

- (i) The system is BIBO stable or internally stable or stable
- (ii) The impulse response function is absolutely integrable, i.e.,

$$\|h\|_1 := \int_0^{\infty} |h(t)| dt < \infty$$

- (iii) All poles of $G(s)$ are in open left half plane (LHP) $\mathbb{C}_-$

Stability Theorem: Proof: (ii) $\Rightarrow$ (i)

Since $y(t) = h(t) * u(t) = \int_0^\infty h(\tau)u(t - \tau)d\tau$ (see Lec. 6 p. 1-7), therefore

$$\begin{aligned} |y(t)| &= \left| \int_0^\infty h(\tau)u(t - \tau)d\tau \right| \\ &\leq \int_0^\infty |h(\tau)u(t - \tau)| d\tau \quad (\text{applying triangle inequality}) \\ &= \int_0^\infty |h(\tau)| |u(t - \tau)| d\tau \\ &\leq \|u\|_\infty \int_0^\infty |h(\tau)| d\tau \end{aligned}$$

So $\|h\|_1 < \infty \Rightarrow \|y\|_\infty < \infty$, showing (ii) $\Rightarrow$ (i)

Stability Theorem: Proof: (i) $\Rightarrow$ (ii)

Contraposition: we will show $\int_0^\infty |h(t)|dt = \infty \Rightarrow$ not BIBO

Consider specific $u(t) := -\text{sign}(h(-t))$. Clearly $\|u\|_\infty = 1 < \infty$, i.e., $u(t)$ is bounded

However, $y(0) = \int_0^\infty h(0 - \tau)u(\tau)d\tau = \int_0^\infty |h(\tau)|d\tau = \infty$, i.e., $y(t)$ is not bounded

This shows (i) $\Rightarrow$ (ii)

Stability Theorem: Proof for (iii) $\Rightarrow$ (ii)

Because the system is causal SISO LTI, the transfer function $G(s)$ is **proper**

Suppose that $G(s)$ has ν **distinct** poles: $p_1, p_2, \dots, p_\nu \in \mathbb{C}$ with multiplicities $n_1, n_2, \dots, n_\nu$ such that $n_1 + n_2 + \dots + n_\nu = n$

Then, by partial fraction expansion, we can write $G(s)$ in residue form (Lec. 8 p. 4)

$$G(s) = k + \sum_{i=1}^{\nu} \sum_{\ell=1}^{n_i} \frac{r_{i\ell}}{(s - p_i)^\ell}$$

where the residues $r_{i\ell} \in \mathbb{C}$, and the **real** constant $k := \lim_{s \rightarrow \infty} G(s) < \infty$. Clearly, $k = 0$ if G is strictly proper, and $\neq 0$ otherwise

Stability Theorem: Proof for (iii) $\Rightarrow$ (ii) continued

Taking $\mathcal{L}^{-1}[\cdot]$ to both sides of the residue form, we get

$$\begin{aligned} h(t) &= \mathcal{L}^{-1}[G(s)] = k\mathcal{L}^{-1}[1] + \sum_{i=1}^{\nu} \sum_{\ell=1}^{n_i} r_{i\ell} \mathcal{L}^{-1} \left[\frac{1}{(s - p_i)^\ell} \right] \\ &= k\delta(t) + \sum_{i=1}^{\nu} \sum_{\ell=1}^{n_i} r_{i\ell} e^{p_i t} \frac{t^{\ell-1}}{(\ell-1)!} \end{aligned}$$

Poles in real-imaginary part form: $p_i = \sigma_i + j\omega_i$ with $\sigma_i, \omega_i \in \mathbb{R}$ for all $i = 1, 2, \dots, n$

Since $|e^{p_i t}| = e^{\sigma_i t}$, by triangle inequality and that $|\delta(t)| = \delta(t)$ (Lec. 5 p. 6), we have

$$|h(t)| \leq |k|\delta(t) + \sum_{i=1}^{\nu} \sum_{\ell=1}^{n_i} |r_{i\ell}| e^{\sigma_i t} \frac{t^{\ell-1}}{(\ell-1)!}$$

Stability Theorem: Proof for (iii) $\Rightarrow$ (ii) continued

For $p_i \in \mathbb{C}_- \Leftrightarrow \sigma_i < 0$, we integrate both sides of the last inequality from $t = 0$ to ∞ :

$$\begin{aligned} \int_0^\infty |h(t)| \, dt &\leq |k| \underbrace{\int_0^\infty \delta(t) \, dt}_{=1 \text{ (Lec. 5 p. 6)}} + \sum_{i=1}^{\nu} \sum_{\ell=1}^{n_i} \frac{|r_{i\ell}|}{(\ell-1)!} \underbrace{\int_0^\infty t^{\ell-1} e^{\sigma_i t} \, dt}_{\substack{\text{Laplace transform of power} \\ = \frac{(\ell-1)!}{(-\sigma_i)^\ell}}} \\ &= |k| + \sum_{i=1}^{\nu} \sum_{\ell=1}^{n_i} \frac{|r_{i\ell}|}{(-\sigma_i)^\ell} < \infty \end{aligned}$$

This shows (iii) $\Rightarrow$ (ii)

Stability Theorem: Proof for (ii) $\Rightarrow$ (iii)

Here, we are given $\int_0^\infty |h(t)| dt < \infty$

For any $s \in \overline{\mathbb{C}}_+$ (closed RHP where $\text{Re}(s) \geq 0$), and for any $t \in [0, \infty)$,

$$|e^{-st}| = e^{-\text{Re}(s)t} \leq 1$$

Therefore, $|G(s)| = \left| \int_0^\infty h(t) e^{-st} dt \right| \leq \int_0^\infty |h(t)| dt < \infty$

This shows (ii) $\Rightarrow$ (iii) since otherwise (i.e., if G had a pole $p_i = \sigma_i + j\omega_i$ with $\sigma_i \geq 0$) $|G(s)| \uparrow \infty$ as $s \rightarrow p_i$, thereby contradicting the bound

Remarks

- The stability theorem and its proof extend to MIMO LTI systems. In SISO case, slide 4 condition (ii) says $\|h\|_1 < \infty$. In MIMO case, arrange the impulse response integrals $\int_0^\infty |h_{ij}(t)|dt$ in a matrix H of size $p \times m$ (number of outputs $\times$ number of inputs). Then condition (ii) becomes matrix 1-norm condition: $\|H\|_1 < \infty$. Condition (iii) generalizes: all poles of $p \times m$ transfer matrix $G(s)$ must lie in $\mathbb{C}_-$
- In slide 3, the notions of BIBO stability and stability/internal stability coincide for LTI systems, but not for general LTV and nonlinear systems
- In slide 4 condition (iii), we tacitly assumed that $G(s)$ has no pole-zero cancellations. If there are unstable pole-zero cancellations (e.g., terms like $(s - 3)$) then the SISO LTI system is BIBO stable in the sense of (i)-(ii), but not internally stable in the sense of (iii). We then say: the SISO LTI realization is non-minimal